

DRY-RUN GUIDE | ss233p.py

Source: Python\3rd_Belt for practice\ss233p.py

CORE BEHAVIOR

Intended to remove a requested substring, but iterates over characters and uses the character as a slice index. Non-empty input therefore raises TypeError before matching.

REPRESENTATIVE INPUT

a='abc', b='b'.

STEP TRACE

First loop value i='a'; evaluating a[i:len(b)] uses a string as a slice bound, so execution stops.

EXPECTED OUTPUT / RETURN

TypeError and no printed result. With empty a, the loop is skipped and an empty string is printed.

ACTUAL SOURCE EXCERPT

Verbatim excerpt; line numbers refer to the original source, which was not modified.

```
001 | # Question: Solve the given array/string challenge using the required logic.
002 | # Add your solution here.
003 |
004 | def compress_string(s):
005 |     if not s:
006 |         return s
007 |     compress = []
008 |     current_char=s[0]
009 |     count=1
010 |     for i in s:
011 |         if s[i]==current_char:
012 |             count+=1
013 |         else:
014 |             compress.append(compress + str(count))
015 |             current_char=s[i]
016 |             count=1
017 |     compress.append(compress + str(count))
018 |     compress_str="".join(compress)
019 |     if len(compress_string) < len(s):
020 |         return compress_string
021 |     else:
022 |         return s
023 | a=input()
024 | b=compress_string(a)
025 | print(b)
026 |
```

DRY-RUN NOTE

Trace describes actual source behavior, including errors where applicable.